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Key Points:

- Artificial intelligence (AI) weather prediction models provide reliable short-term forecasts of major heatwaves (HWs) across diverse climate regions
- A hybrid method that combines AI weather forecasts with climate change estimates from global models enables rapid attribution of HWs
- Sea surface temperature forcing is secondary to atmospheric initial conditions in driving HW attribution signals

Supporting Information:

Supporting Information may be found in the online version of this article.

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AI-Driven Weather Forecasts to Accelerate Climate Change Attribution of Heatwaves

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Abstract Anthropogenic climate change (ACC) is driving an increase in the frequency, intensity, and duration of heatwaves (HWs), making the rapid attribution of these events essential for assessing climate-related risks. Traditional attribution methods often suffer from selection bias, high computational costs, and delayed results, limiting their utility for real-time decision-making. In this study, we introduce a novel artificial intelligence (AI)-driven attribution framework that integrates physics-based ACC estimates from global climate models with state-of-the-art AI weather prediction (AIWP) models. We apply this approach to four HWs across different climatic regions using two AIWP models (FourCastNet-v2 and Pangu-Weather) and one hybrid AI-physics model (NeuralGCM). Our results show that AIWP models accurately predict HW intensity and spatial patterns, capturing key synoptic features such as persistent high-pressure ridges. The attribution analysis reveals a robust ACC signal in all four events and a good agreement across models. Results from the hybrid model (NeuralGCM) suggest that the intensification of HWs due to ACC can largely be inferred from the atmospheric state a few days prior to the event, while sea surface temperature forcing becomes increasingly relevant at longer lead times and in specific regions. This study demonstrates that AI-based attribution enables near real-time and anticipatory assessment of HWs, offering a scalable and computationally efficient alternative to conventional methods. By providing timely and consistent attribution of extreme heat events, this approach enhances our ability to anticipate climate risks and inform adaptation strategies in a rapidly warming world.

Plain Language Summary It is well accepted that climate change is increasing the occurrence and severity of heatwaves (HWs). However, when an event happens, it is important to understand how much climate change influenced that event. Traditional methods are often slow or require expensive resources, making them less useful for quick decision-making. This study introduces a new approach using weather models powered by artificial intelligence (AI) to quickly estimate how climate change affects the intensity and characteristics of HWs. The method combines fast AI predictions with data from global climate models. We apply this approach to four major HWs in various climate regions and show that AI models can accurately predict HW conditions, including features like persistent high-pressure systems. Our analysis finds that climate change made all four HWs more intense than they would have been in a preindustrial climate, with strong agreement across models. The results also show that while sea surface temperatures play a role, the main influence comes from changes in atmospheric conditions. This AI-based method enables faster, near real-time attribution of HWs, supporting quicker decisions and better planning in a warming world.

1. Introduction

Extreme weather events pose significant risks to ecosystems and society (Ummenhofer & Meehl, 2017). Among them, heatwaves (HWs) have become one of the most devastating climate-related hazards, with anthropogenic climate change (ACC) contributing to their increasing frequency, severity, and duration across many regions (Seneviratne et al., 2021). These trends are expected to intensify, potentially causing irreversible damage to ecosystems and escalating threats to human health, infrastructure, and economies (Kumar et al., 2023). The ability to understand, predict, and attribute such extremes is therefore more critical than ever.

Attributing HWs to ACC is essential for raising public awareness of climate impacts and providing a scientific foundation for adaptation strategies, mitigation policies, risk management and decision-making on loss and damage (Coumou et al., 2024; Knutson et al., 2017; NAS, 2016; Otto, 2023; Perkins-Kirkpatrick, Alexander,

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et al., 2024; Stott et al., 2016). Improving our ability to predict and attribute HWs in near real-time is crucial for anticipating associated risks and enhancing societal resilience to future extremes. However, despite advances in attribution science (Athanas et al., 2024; Faranda et al., 2024; Philip et al., 2020; Stott & Christidis, 2023), existing methods often struggle to deliver timely attribution while an event is unfolding—precisely when such information is most demanded. Furthermore, traditional attribution approaches are typically constrained by predefined event definitions and case selection, limiting their generalization and explicit attribution of ACC's influences on the spatio-temporal patterns of the event.

Artificial intelligence weather prediction (AIWP) models offer an unprecedented opportunity to address these limitations. By generating global forecasts in minutes on standard hardware and with a skill that rivals or even exceeds traditional numerical weather prediction models (NWP) (Olivetti & Messori, 2024; Rasp et al., 2023), AIWP models can enable near-instantaneous attribution analyses. State-of-the-art AIWP models such as Pangu-Weather (Bi et al., 2023), FourCastNet (Pathak et al., 2022), FourCastNet-v2 (Bonev et al., 2023) or GraphCast (Lam et al., 2023) have already demonstrated strong capabilities in forecasting extreme events, such as HWs, extratropical storms, or tropical cyclones (Bouallège et al., 2024; Charlton-Perez et al., 2024; Pasche et al., 2025). Trained on historical reanalysis data, these models learn representations of atmospheric dynamics and can produce forecasts that evolve in physically consistent ways, including the formation of baroclinic waves, tropical cyclones, and frontal weather systems (Baño-Medina et al., 2025; Charlton-Perez et al., 2024; Hakim & Masanam, 2024).

This study introduces an innovative, operationalizable approach for ultra-fast ACC attribution based on AI-based weather forecasting, which enables attribution before an event occurs. As in classical attribution, the event is simulated in a “factual” world where it occurred, and in a hypothetical “counterfactual” world without ACC (Ciavarella et al., 2018; Jézéquel et al., 2018; Philip et al., 2020). The ACC signal, derived from global climate models (GCMs) and the pseudo-global warming (PGW) technique (Brogli et al., 2023), is subtracted from the observed pre-event conditions, allowing AIWP model forecasts to estimate how the event would have evolved in a world without human-induced climate change.

This framework aligns with highly constrained storyline attribution methodologies (Shepherd, 2016; van Garderen et al., 2021), which offer physically plausible, event-specific narratives of how climate change has altered the characteristics of specific extremes. These methods have been applied to a range of hazards—including HWs (Bercos-Hickey et al., 2022; Klimiuk et al., 2025; Leach et al., 2021, 2024; Sánchez-Benítez et al., 2022), convective systems (González-Alemán et al., 2023; Martín et al., 2024), tropical cyclones (Patricola & Wehner, 2018; Reed et al., 2020; Wang et al., 2024), and extratropical cyclones (Ermis et al., 2024). However, previous studies have relied on global (sometimes regional) NWP simulations, which are computationally expensive, and require high-performance computing and expert knowledge, barriers that limit real-time global attribution. By contrast, our framework would enable near-instantaneous attribution of events. The method is illustrated for set of recent HWs with varying characteristics that impacted different regions worldwide.

In this study, we use two AIWP models, FourCastNet-v2 (Bonev et al., 2023) and Pangu-Weather (Bi et al., 2023), along with one hybrid model, NeuralGCM (Kochkov et al., 2024). While FourCastNet-v2 and Pangu-Weather are purely data-driven models that learn atmospheric dynamics from reanalysis data, NeuralGCM integrates a differentiable solver for atmospheric motion with AI-based parameterizations for unresolved physics. By incorporating prescribed sea surface temperature (SST) and sea ice as boundary conditions, NeuralGCM also accounts for external oceanic influences rather than inferring them implicitly from the training data. A recent study by Duan et al. (2025) applied NeuralGCM to the 2021 Pacific Northwest HW, demonstrating its potential for attribution applications, while also highlighting limitations due to missing land–surface feedbacks. Our work extends this line of research by systematically evaluating multiple AIWP models across diverse HW events, allowing us to assess both the strengths and current limitations for forecast-based attribution under a range of synoptic and surface conditions. The use of these models in our study allows us to assess: (a) how purely AI-driven forecasts compare to those of a model that explicitly simulates atmospheric dynamics and (b) the robustness of forecast-based attribution when accounting for ACC effects on boundary conditions—an aspect typically absent in most data-driven AIWP models.

Table 1
Characteristics of the Models Employed in the Study

	FourCastNet-v2	Pangu-weather	NeuralGCM
Architecture	Spherical Fourier Neural Operators (SFNO)	3D Transformer	AI physics (CNN) + dynamical core differential solver
Training period	1979–2015	1979–2017	1980–2017
Resolution	0.25°—6 hourly	0.25°—6 hourly	1.4°—6 hourly
Model levels	Thirteen pressure levels (1,000, 925, 850, 700, 600, 500, 400, 300, 250, 200, 150, 100, and 50) hPa	Thirteen pressure levels (1,000, 925, 850, 700, 600, 500, 400, 300, 250, 200, 150, 100, and 50) hPa	Thirty-two hybrid sigma-pressure levels (interpolated from the 37 pressure levels of ERA5)
Variables	Five pressure-level (u, v, z, t , r) and 8 surface (10u, 10v, 100u, 100v, msl, sp, 2t , tcwv)	Five pressure-level (u, v, z, t , q) and 4 surface (10u, 10v, msl, 2t)	Six pressure-level (u, v, z, t , q , ciwc, clwc), 2 forcings (sst , siconc)

Note. Variable names correspond to the short names as defined by the GRIB2 convention. Variables in Bold are those modified to compute counterfactual forecasts.

2. Methods

2.1. AIWP Models and Heatwave Forecasts

We use three models in this study: two AIWP models, FourCastNet-v2 (Bonev et al., 2023) and Pangu-Weather (Bi et al., 2023), along with NeuralGCM (Kochkov et al., 2024), a hybrid AI-physics model. Further details on these models are provided in Table 1 and the following paragraphs.

2.1.1. FourCastNet-v2

FourCastNet-v2, an improved version of FourCastNet (Pathak et al., 2022), developed by Nvidia and collaborators, provides global predictions at 0.25° horizontal resolution for 13 pressure levels at 6-hr intervals (see Table 1). It forecasts a range of variables, including five pressure-level variables (temperature, zonal wind, meridional wind, geopotential height, and relative humidity) and eight surface variables (2m temperature, sea level pressure, surface pressure, vertically integrated water vapor, and wind components at 10 and 100 m). The model achieves comparable skill to the ECMWF Integrated Forecasting System (IFS) for large-scale variables (European Centre for Medium-Range Weather Forecasts, 2025). FourCastNet-v2 builds upon its predecessor by incorporating Spherical Fourier Neural Operators (SFNOs), which enhance forecast accuracy by ensuring physical consistency with Earth's spherical geometry (Bonev et al., 2023). It is trained in two stages: an autoregressive single-step training (6-hr predictions), followed by fine-tuning with multiple autoregressive steps.

2.1.2. Pangu-Weather

Pangu-Weather, developed by Huawei Cloud AI, is another AI-based global forecasting model that employs a three-dimensional Earth-specific Transformer architecture (Bi et al., 2023). It provides global predictions at a 0.25° horizontal resolution for 13 pressure levels at 6-hr intervals (see Table 1). The model forecasts a range of variables, including five pressure-level variables (temperature, zonal wind, meridional wind, geopotential height, and specific humidity) and four surface variables (2 m temperature, sea level pressure, and wind components at 10 m). To generate forecasts, Pangu-Weather uses a hierarchical time-stepping approach, recursively predicting future states and combining models trained at different lead times to improve stability and accuracy over longer horizons. Pangu-Weather has demonstrated high accuracy in predicting extreme weather events such as hurricanes and temperature extremes (Bouall  gue et al., 2024).

2.1.3. NeuralGCM

NeuralGCM is a machine-learning-based general circulation model (GCM) that integrates neural network parameterizations with classical dynamical core formulations (Kochkov et al., 2024). This hybrid approach combines the strengths of traditional physics-based models with the speed and pattern recognition capabilities of deep learning. NeuralGCM is designed to maintain physical consistency, particularly in energy conservation and wave propagation—two key challenges for purely AI-based models. It achieves competitive skill scores for medium-

Table 2
Coupled Model Intercomparison Project Phase 6 Models Used for Calculating the Climate Change Deltas of Thermodynamic Variables

Model acronym	Name	Reference
ec-earth3-veg-lr	EC-Earth3—Vegetation—Low Resolution	Döscher et al. (2022)
ec-earth3-cc	EC-Earth3—Carbon Cycle	Döscher et al. (2022)
ec-earth3-veg	EC-Earth3—Vegetation	Döscher et al. (2022)
awi-cm-1-1-mr	Alfred Wegener Institute—Climate Model—Medium Resolution	Semmler et al. (2020)
canesm5-1	Canadian Earth System Model version 5	Swart et al. (2019)
cmcc-cm2-sr5	CMCC—Climate Model 2—System Resolution 5	Lovato et al. (2022)
cmcc-esm2	CMCC—Earth System Model 2	Lovato et al. (2022)
cmcc-cm2-hr4	CMCC—Climate Model 2—High Resolution 4	Scoccimarro et al. (2019)
bcc-csm2-mr	Beijing Climate Center—Climate System Model 2—Medium Resolution	Xin et al. (2018)
cams-csm1-0	Chinese Academy of Meteorological Sciences Climate System Model 1.0	Rong (2019)

range weather forecasting and, unlike other AIWP models that lose sharpness over time, maintains a nearly constant spectral structure throughout the forecast period.

In this study, we use the deterministic configuration of NeuralGCM with a horizontal resolution of 1.4° , employing 37 vertical pressure levels. The model predicts a set of key atmospheric variables, including temperature, zonal and meridional winds, geopotential height, specific humidity, cloud liquid water content, and cloud ice water content. Sea surface temperatures (SSTs) and sea ice cover are prescribed from the initial state and remain constant throughout the 10-day simulation, following the typical setup for NeuralGCM in weather forecasting mode. However, prescribing SSTs as fixed boundary conditions prevents two-way ocean–atmosphere interactions, potentially underestimating the role of feedbacks in modulating HW intensity and persistence, particularly at longer timescales (beyond 1 week).

2.1.4. Forecasting and Model Initialization

All three models—FourCastNet-v2, Pangu-Weather, and NeuralGCM—were trained on the ERA5 reanalysis data set (Hersbach et al. (2020), see Table 1 for details). For each HW event, we generate 10-day 6-hourly forecasts at multiple lead times, starting several days before the event and continuing until its conclusion. For factual simulations, the models are initialized from ERA5 reanalysis, and we generate a lagged ensemble of 16 members comprising all lead times ranging from 1 to 5 days. This approach follows the methodology of Brenowitz et al. (2024) and allows us to compute an ensemble mean while assessing the uncertainty associated with model initialization. As demonstrated in Brenowitz et al. (2024) for the IFS model, lagged ensembles provide a reliable estimate of forecast skill and uncertainty, aligning well with operational ensemble predictions.

The 1–5 days lead time window is chosen to balance two needs: maintaining high forecast accuracy and fidelity to the observed synoptic evolution, while allowing enough integration time in the counterfactual simulations for the imposed thermodynamic perturbation to influence the forecast dynamically. These short lead times help constrain the circulation to the observed event, and enable the simulated outcome to adjust to the perturbation.

To diagnose HW's magnitude, we use daily mean temperature at 850 hPa (T850), which is forecasted by the three models.

2.2. Climate Change Attribution Method

The method removes the ACC signal from the initial conditions of the factual simulations using the PGW approach (Brogli et al., 2023). First, monthly averages for various thermodynamic variables are derived from historical (1850–2014) simulations of 10 Coupled Model Intercomparison Project Phase 6 (CMIP6, Eyring et al., 2016) GCMs (see Table 2). For each GCM, only one realization is used (r1i1p1f1). The considered variables

depend on the AIWP model employed (see variables in bold in Table 1). These variables include 2 m temperature (2D, 2t), total column water vapor (2D, tcwv), air relative humidity (3D, r), air specific humidity (3D, q), and air temperature (3D, t). Additionally, we also consider SST (tos) and sea ice concentrations (siconc) for the NeuralGCM model. To maintain hydrostatic balance, the geopotential height is adjusted in accordance with the modified variables. All CMIP6 data are interpolated into a common regular $2.5 \times 2.5^\circ$ grid. We selected 10 CMIP6 models that provided all the necessary variables and included interpolated values below topography for pressure levels—a requirement, as AIWP models cannot process missing input data. Models containing missing values were excluded. Since the goal of this study is not to precisely quantify the climate change impact on individual events, but rather to demonstrate the feasibility and robustness of the new framework, the specific choice of CMIP6 models is not critical to our results.

For each thermodynamic variable X , with $X = (2t, tcwv, t, r, q)$, the ACC signal (ΔX_{ACC}) is defined as the difference between the factual (1980–2014) and counterfactual (1850–1900) climatologies:

$$\Delta X_{ACC}(g, m, x, y, z) = \overline{X}_{1980-2014}(g, m, x, y, z) - \overline{X}_{1850-1900}(g, m, x, y, z) \quad (1)$$

Here, the overbar denotes the multi-year mean, g the CMIP6 model, m the calendar month of the climatology, (x, y) the grid point, and z the pressure level. The multi-model mean (MM) ACC signal is then computed for each thermodynamic variable X . The obtained MM corresponds to a warming level (global area-averaged surface temperature) of 0.85°C , calculated as the difference between the 1980–2014 climatology and the preindustrial baseline (1850–1900). This warming level is below current observational estimates ($\sim 1.3^\circ\text{C}$); therefore, the results should not be interpreted as precise quantifications of anthropogenic influence, but rather as a demonstration of the method's applicability across different AIWP models.

Finally, the monthly $\Delta X_{ACC,MM}(m, x, y, z)$ is linearly interpolated to the exact initialization date of the counterfactual simulations (by taking the monthly means as representative of the 15th day of each month), and also linearly interpolated to the ERA5 grid (0.25°). For NeuralGCM simulations, the pressure-level variables are additionally interpolated to the 37 ERA5 pressure levels used by the model.

The initial conditions of the counterfactual simulations are obtained by subtracting the thermodynamic perturbations from the initial conditions of the factual runs ($X' = X - \Delta X_{ACC}$), and adjusting the geopotential height to be in hydrostatic balance with the introduced perturbations. The geopotential height perturbation ($\Delta\phi_{ACC}$) is calculated as the difference between the hydrostatically adjusted geopotential of the factual conditions (ERA5) and counterfactual (X') conditions. To compute the hydrostatic geopotential, we integrate pressure levels, from bottom to top, using the following equation (based on Equation 6 in Brogli et al. (2023)):

$$\phi_{z+1} = \phi_z + (\ln(p_z) + \ln(p_{z+1})) R \frac{(t_{v,z} + t_{v,z+1})}{2} \quad (2)$$

where $z = 1, \dots, 13$ is the index of model pressure-level (where $z = 1$ corresponds to $p_1 = 1000\text{hPa}$), $R = 287\text{Jkg}^{-1}\text{K}^{-1}$ the ideal gas constant, and $t_{v,z}$ the virtual temperature at pressure level z , computed using t_z and r_z or q_z depending on the model humidity variable. The $\Delta\phi_{ACC}$ is then subtracted from the ERA5 initial conditions before running the counterfactual simulations. We adjust the geopotential in the counterfactual initial conditions only for FourCastNet-v2 and Pangu-Weather, as the hybrid model (NeuralGCM) more easily adapts to the initial temperature perturbation. Tests comparing NeuralGCM simulations with and without the geopotential adjustment show no difference in the attribution results. Figure S1 in Supporting Information S1 displays the calculated ACC deltas for a specific initialization date.

Once the initial counterfactual conditions are obtained, we repeat the same procedure as used for the factual simulations. This includes generating a lag-based ensemble of counterfactual forecasts for the same initialization dates and lead times as the factual runs. The ACC signal on HW's magnitude is then defined as the difference between the lagged ensemble mean of the factual and counterfactual simulations. Figure 1 provides a schematic overview of the attribution method.

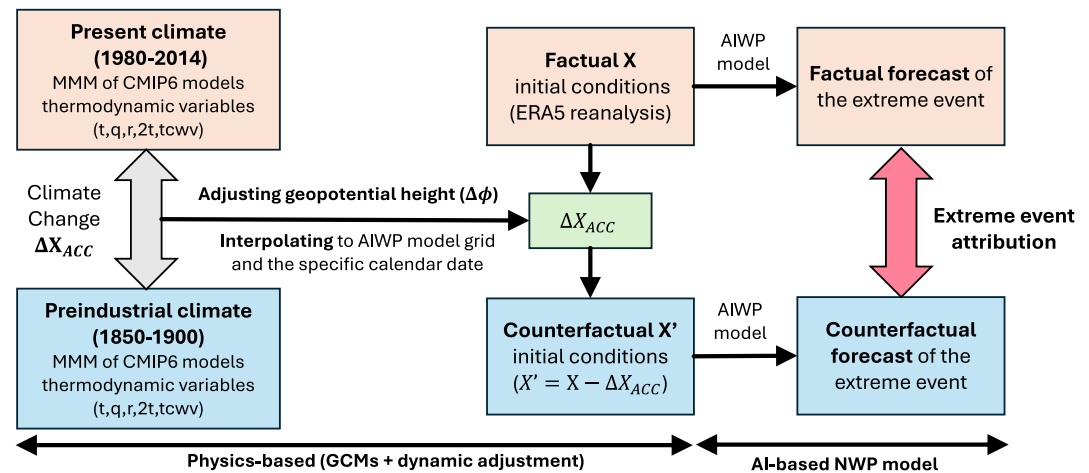


Figure 1. Schematic representing the steps involved in the attribution method.

2.3. Statistical Significance of the Results

To assess the statistical significance of the ACC attribution signal in the 2D T850 fields, we employ a two-sided t -test using the `ttest_ind` function from the `scipy` Python package (Virtanen et al., 2020). This test assumes that the two distributions of the factual and counterfactual worlds have equal variances and are independent of each other. Using a t -test with two independent variances yields similar results. For each predicted 6-hourly time step, the distribution's population contains the 16 forecasts initialized from 24 to 120 hr before.

3. Results

We analyze the model forecast skill and ACC attribution for four recent outstanding HWs that hit different regions of the two hemispheres. These events were chosen to capture a broad geographic diversity, allowing us to assess model performance across different climatic zones. Importantly, all events fall outside the models' training period (1979–2017), providing a robust test of the AIWP models' capabilities for generalization. These events include the Pacific Northwest 2021 HW (28–30 June 2021, e.g., Leach et al., 2024; Schumacher et al., 2022; Thompson et al., 2022), the Iberian 2018 HW (1–6 August 2018, e.g., Barriopedro et al., 2020), the Brazilian 2023 HW (12–18 November 2023, e.g., Collazo et al., 2025; Messori & Faranda, 2024; Perkins-Kirkpatrick, Barriopedro, et al., 2024), and the India-Pakistan 2022 HW (14–18 March 2022, e.g., Kew et al., 2023; Zachariah et al., 2023). The event dates are selected from the aforementioned studies (for the India-Pakistan HW, we consider the first 5 days of the HW, as it lasted almost 1 month). These HWs have varying duration, intensity and nature (e.g., humid vs. dry events) and span contrasting regions (subtropical vs. extratropical, coastal vs. continental). Moreover, the considered events have been attributed to climate change using a diversity of methods, including NWP forecast-based approaches as the one employed here with AIWP models.

3.1. Forecast Skill

For each HW, we first evaluate the ability of the two AIWP models and the hybrid model to predict the observed magnitude and spatial pattern of the geopotential at 500 hPa (Z500) and temperature at 850 hPa (T850) fields in the ERA5 reanalysis data set (Hersbach et al., 2020). We use T850 because it is provided by the three models (temperature at 2 m, T2m, is only available in two models). Note that the three models have been initialized from ERA5, which is also the data set used to train their AI algorithms (see Methods). The bias is computed as the difference between the model forecasts and ERA5 for each day and forecast lead time.

The four HWs under study are characterized by high geopotential heights and the formation of atmospheric ridges (Figures 2a, 2d, 2g, and 2j). These ridges create stable atmospheric conditions that favor subsidence, reduced cloud cover, and increased solar radiation, all of which contribute to rising surface and near-surface temperatures (Barriopedro et al., 2023; Domeisen et al., 2023). Additionally, land-atmosphere feedbacks, such as reduced evapotranspiration due to drought or dry soil conditions, can further intensify these HWs. While these feedbacks

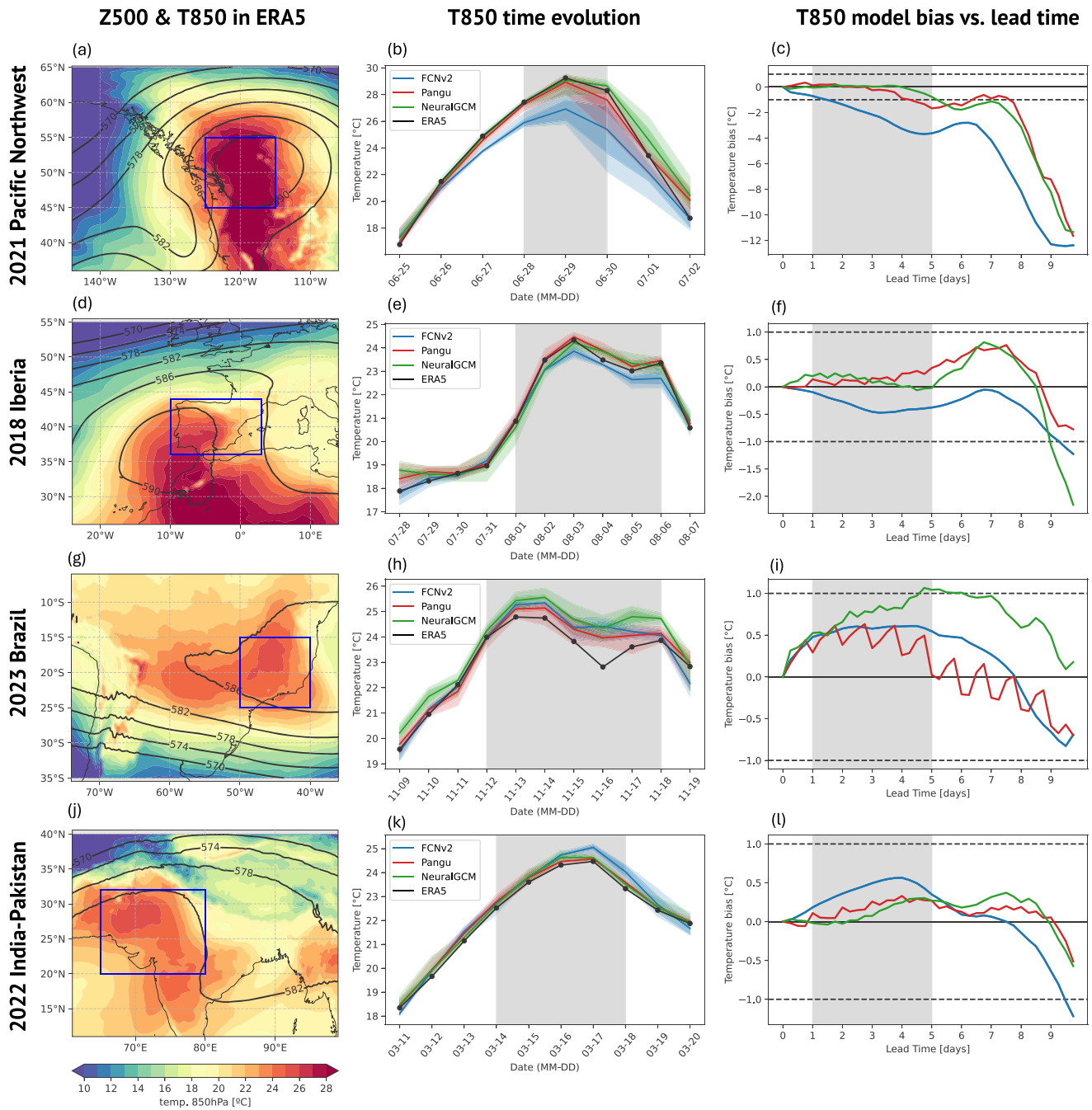


Figure 2. Artificial intelligence-based forecasts for four outstanding heatwaves (HWs). (a–c) 2021 Pacific Northwest HW; (d–f) 2018 Iberian HW; (g–i) 2023 Brazilian HW; and (j–l) 2022 India-Pakistan HW. (a, d, g, and j) Daily mean temperature at 850 hPa (T850, shading, in °C) and geopotential height at 500 hPa (Z500, black contours, in dam) averaged over the duration of the HW, based on ERA5 reanalysis data. (b, e, h, and k) Temporal evolution of area-weighted daily mean T850 (°C) over the blue box shown in (a, d, g, and j) for each model (see legend) and ERA5. Light shading represents the maximum and minimum values of the lagged ensemble (lead times of 1–5 days), while dark shading shows its interquartile range. The HW period is highlighted in gray; (c, f, i, and l) Model bias in T850 (°C) as a function of lead time, averaged over the HW period. The lead time interval of 1–5 days used to compute the lagged ensemble mean in panels (b, e, h, and k) is shaded in gray.

are not explicitly simulated in any of the models used in this study, it is possible that some of these nonlinear physical processes were indirectly encoded during the model training phases.

Overall, the three models perform reasonably well in predicting the four HWs, with small temperature biases for forecast lead times between 1 and 5 days, the period used to compute the lagged ensemble mean (Figures 2b, 2c,

2h, and 2k). During the peak days of each event, the models generally exhibit good performance, with low bias compared to ERA5. Among the four events, the 2021 Pacific Northwest HW shows the largest underestimation, particularly in FourCastNet-v2 (Figure 2b). This bias worsens with the forecast lead time (Figure 2c), reaching nearly -4°C at 4–5 days lead time when averaged over the HW region (blue box in Figure 2a). The other two models also underestimate the event's magnitude, but at longer lead times, with NeuralGCM showing a small bias for forecasts shorter than 5 days. Notably, all three models fail to capture this HW when initialized more than 7–8 days in advance (Figure 2c), suggesting low intrinsic predictability of the event—likely due to a cascade of remote drivers—as shown by simulations with a traditional NWP model in Oertel et al. (2023). The superior performance of the hybrid model for the Pacific Northwest event suggests that this HW was unprecedented from a dynamic perspective, highlighting the advantage of numerically simulating atmospheric flow in such cases. These results are consistent with Pasche et al. (2025), who also found that Pangu-Weather outperforms FourCastNet, demonstrating skill comparable to the IFS-HRES numerical weather prediction (NWP) model for this particular event.

For the 2018 Iberian HW, FourCastNet-v2 slightly underestimates temperatures ($\sim 0.5^{\circ}\text{C}$), while Pangu-Weather tends to overestimate them slightly. NeuralGCM, on the other hand, shows the smallest bias for lead times below 5 days (Figures 2e and 2f). During the 2023 Brazilian HW, all models slightly overestimate temperatures, with NeuralGCM showing the largest bias (around 1°C at a 5-day lead time; Figures 2h and 2i). Finally, for the 2022 India-Pakistan HW, all models exhibit a small positive bias, less than 0.5°C , with NeuralGCM performing the best among the three models (Figures 2k and 2l). These results suggest that the three models do not exhibit a consistent systematic bias and are generally effective at capturing the intensity of diverse HWs, despite their differing regional contexts and characteristics. For example, the Brazilian HW was marked by extreme humidity (Perkins-Kirkpatrick, Barriopedro, et al., 2024), whereas the Iberian event unfolded under very dry atmospheric conditions. Interestingly, T850 tends to be slightly overestimated for subtropical HWs (Brazil and India-Pakistan), while it is generally underestimated for extratropical summer HWs (Pacific Northwest and Iberia). This likely reflects the distinct physical mechanisms driving these events: extratropical HWs are often shorter-lived and dynamically driven by large-scale circulation anomalies, whereas subtropical HWs typically evolve more slowly (Jiménez-Estève & Domeisen, 2022).

When examining how model error evolves with forecast lead time (right column in Figure 2 and spatial patterns in Figures S2–S5 in Supporting Information S1), a striking contrast emerges between the predictability of the Pacific Northwest HW and the other three events. For instance, at a 9-day lead time, FourCastNet-v2 underestimates the Pacific Northwest HW by approximately 12°C , whereas for the other events, the underestimation remains below 1°C . This discrepancy may stem from the fact that the Pacific Northwest HW was a truly unprecedented event (Thompson et al., 2022), with no historical analog in the training data of the AI models, unlike the other regions, which may include similar-intensity events such as the 2003 Iberian HW. Another possible explanation is the inherently lower predictability of the Pacific Northwest event, which was driven by a complex interplay of dynamic and thermodynamic mechanisms (Schumacher et al., 2022) and influenced by remote drivers that introduced a “predictability barrier” (Oertel et al., 2023).

When examining the spatial bias patterns across the models (Figures S2–S5 in Supporting Information S1), a clear correlation can be identified between the biases in Z500 and T850. This relationship is especially pronounced in the Pacific Northwest HW, where the underestimation of the blocking high-pressure system's strength leads to a negative bias in T850 (Figure S2 in Supporting Information S1). This is also seen for the Iberian HW (Figure S3 in Supporting Information S1), though the biases are weaker compared to the Pacific Northwest event. The connection between the Z500 and T850 biases is less evident for the subtropical HWs of Brazil (Figure S4 in Supporting Information S1) and India-Pakistan (Figure S5 in Supporting Information S1), where the Z500 pattern is accurately predicted (low bias). This highlights the importance of accurately predicting the synoptic circulation to forecast HW intensity, at least in midlatitudes (Barriopedro et al., 2023).

In summary, among the three models, NeuralGCM and Pangu-Weather tend to deliver the most accurate HW intensity simulations and have the longest prediction horizon. However, we do not identify a systematic model behavior across HW events, and the physics-informed model is not always the best-performing one. These results also indicate that SST forcing may not play a significant role at the lead times considered, as NeuralGCM (a hybrid model with prescribed SST) and Pangu-Weather (a pure AIWP model relying solely on initial conditions) perform similarly for the four HWs analyzed.

3.2. Attribution of Heatwaves Using AI Weather Models

In this section, we analyze the predicted ACC signal for the four HWs. The ACC signal is calculated as the difference between the factual simulations (initialized with ERA5 initial conditions) and the counterfactual simulations, which are initialized using modified ERA5 initial conditions that remove the ACC signal in thermodynamic variables (see Methods for details). The goal is not to accurately quantify the influence of ACC on these HWs, but rather to demonstrate the applicability of the attribution approach to different AIWP models and to evaluate their degree of agreement and consistency with the results obtained from hybrid physics-based forecasts.

Figure 3 shows the geographic distribution of the ACC signal on T850, averaged for all days of each HW event and the lagged ensemble mean. In general, ACC fingerprints look similar across the three models, for each of the analyzed cases. As expected from the modified initial conditions, all the models predict an intensification of the heat extremes. To characterize the evolution of the ACC signal, we decomposed it into a thermodynamic component (the modified initial conditions) and the dynamically integrated component—defined as the residual after removing the initial perturbation (see Figure S6 in Supporting Information S1). This allows us to quantify how much of the total signal is modulated by the atmospheric evolution constrained by the AIWP models. The magnitude and temporal evolution of these signals are further examined in Figure 4. Of all the events, the Iberian HW of 2018 is the one predicted to strengthen more due to ACC (between 1.3 and 1.4°C, depending on the model, Figure 4d). The predicted warming pattern exhibits a pronounced amplification beneath the atmospheric ridge responsible for the HW (Figures 3d–3f). A similarly localized warming signal is observed for the 2023 Brazilian HW (Figures 3g–3i), whereas the signal is less distinct for the 2022 India–Pakistan (Figures 3j–3l) and 2021 Pacific Northwest events (Figures 3a–3c). This would suggest that local feedbacks amplifying the ACC signal were less important in the latter cases. The evolution of the ACC signal with lead time is shown in Figures S7–S10 in Supporting Information S1 for each event. As lead time increases, the attributable patterns become more complex, and differences across AIWP models grow due to the diverging forecast trajectories.

The ACC signal for the 2021 Pacific Northwest HW presents a more complex pattern, characterized by enhanced warming to the northwest of the ridge and a non-significant area of weaker warming near coastal regions (Figures 3a–3c). All three models predict this coastal pattern, although with some variations in the precise location of the area with low non-significant ACC influence. The forecast bias over that region displays a similar dipolar structure to the ACC pattern (Figure S2 in Supporting Information S1), suggesting that the reduced ACC signal may be linked to increased errors in that area. This pattern in the ACC signal develops as errors grow (particularly around day 3 of the forecast, Figure S7 in Supporting Information S1). By a 5-day lead time, the spatial pattern of ACC signals diverge noticeably across the three models (Figure S7 in Supporting Information S1), which is a consequence of the low predictability of this event. This forecasting bias may also contribute to increased uncertainty in the attributable temperature, which tends to reduce the ACC signal (see, e.g., 30 June in Figures 4a and 4b).

The ACC signal also exhibits day-to-day variations (Figures 4a, 4c, 4e, and 4g), with the 2018 Iberian HW showing the largest ACC signal during its peak days (Figure 4c). This is likely due to local feedback mechanisms associated with anticyclonic conditions, which would locally amplify the HW intensity. While the underlying drivers may be difficult to disentangle, the results demonstrate that the evolving ACC patterns are not merely a linear response to the imposed initial condition perturbations. Instead, they reflect the added value of simulating the evolution of the atmospheric conditions in shaping the climate change signal at the event scale. The fact that a large part of the total ACC signal is already present in the initial atmospheric perturbation—derived from global GCMs (Figure S6 in Supporting Information S1)—is consistent with the overall consensus that ACC-induced HW effects are primarily thermodynamically driven (Barriopedro et al., 2023; NAS, 2016). The event-specific dynamical evolution, as captured by the AIWP models, then serves to modulate, enhance, or reduce this thermodynamic baseline. While this modulation is clearly important for refining the attribution signal in HWs, it is likely to become even more critical for events primarily governed by circulation changes, such as extratropical cyclones.

Overall, NeuralGCM predicts the strongest ACC signals for the four events, while FourCastNet-v2 predicts the weakest, with Pangu-Weather generally producing results closer to FourCastNet-v2, except for the case of the India-Pakistan HW (Figure 4h). However, it does not necessarily mean an underperformance of the data-driven models. Indeed, we do not detect a systematic behavior across HWs in any of the models. A notable exception is the India-Pakistan HW, where FourCastNet-v2 predicts a much weaker ACC signal compared to the other two

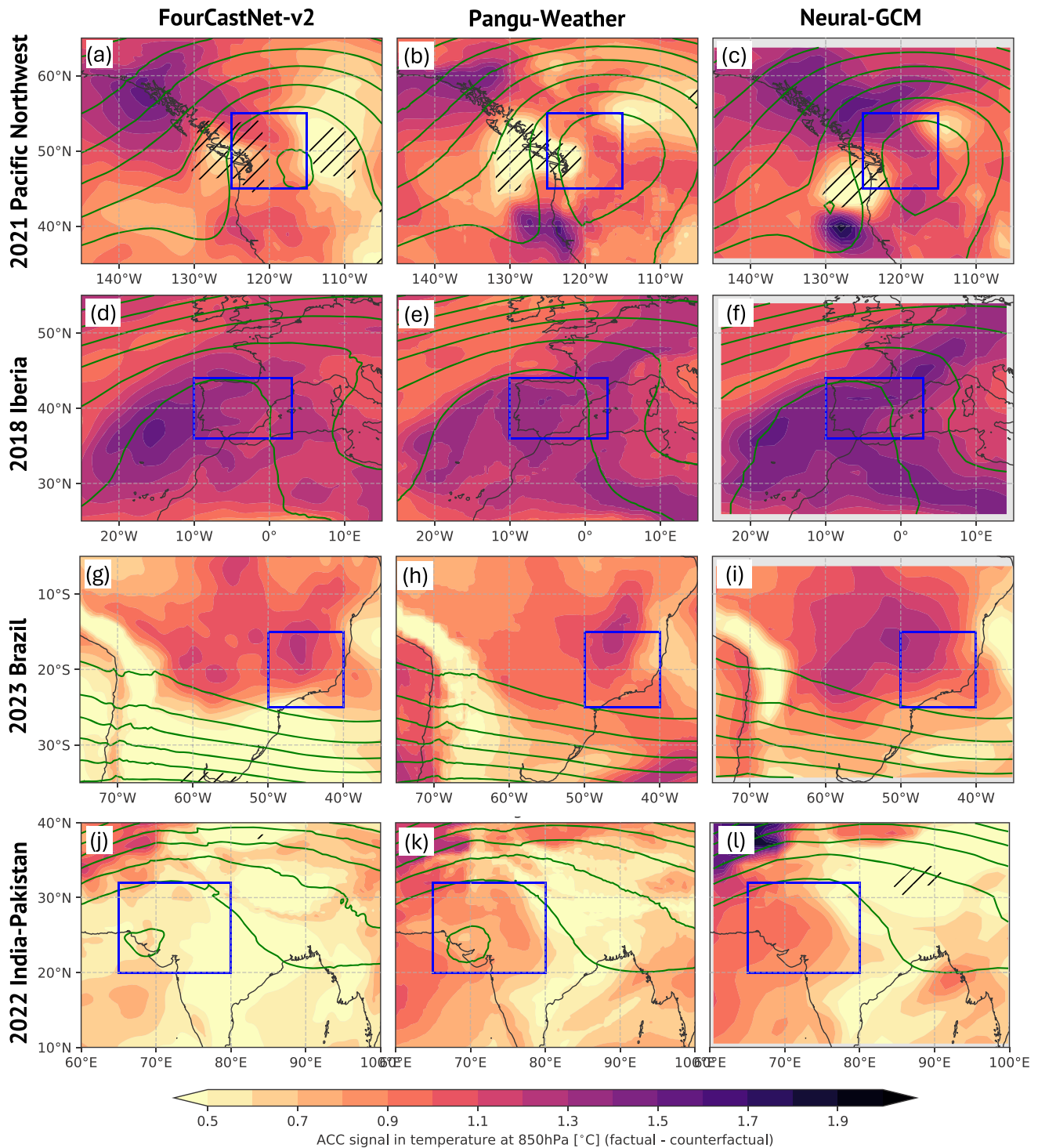


Figure 3. Artificial intelligence-based attribution of heatwaves (HWs) to Anthropogenic climate change (ACC). (a–c) 2021 Pacific Northwest HW; (d–f) 2018 Iberian HW; (g–i) 2023 Brazilian HW; and (j–l) 2022 India-Pakistan HW. ACC signal (factual minus counterfactual difference of lagged ensemble means) in T850 (°C) for each HW (rows), as simulated by the three models (columns): (a, d, g, and j) FourCastNet-v2; (b, e, h, and k) Pangu-Weather; and (c, f, i, and l) NeuralGCM. Z500 (contour interval of 40 dam) forecasts in the factual simulations are represented as green contours for interpretability. Non-significant differences at the 95% confidence level according to a two-sided *t*-test are hatched. Blue boxes are the HW regions used for the computation of area-weighted means.

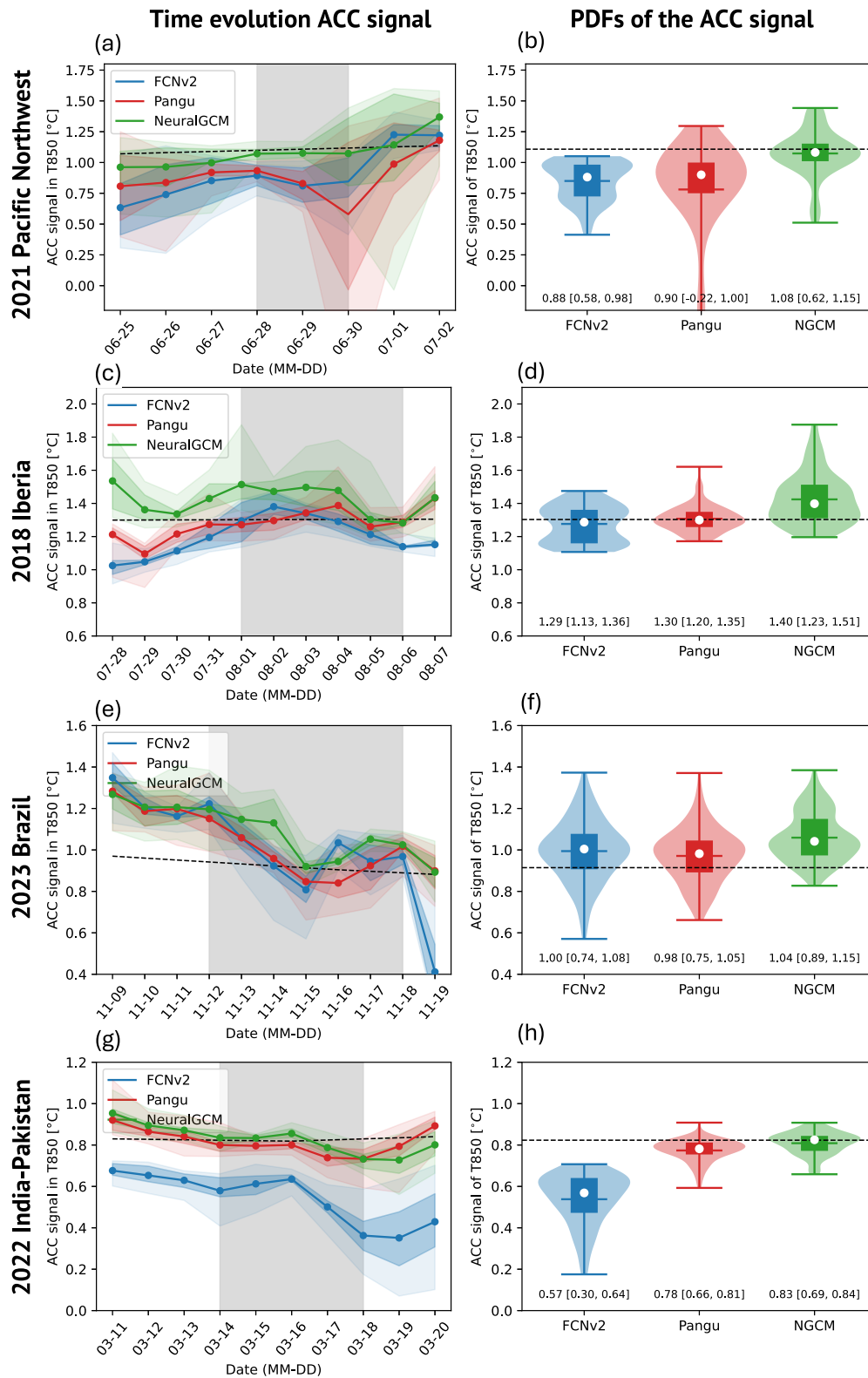


Figure 4.

models (Figures 4g and 4h). Despite these differences, the ACC signals predicted by the models show a large degree of consistency, with significant overlap in their distributions (Figures 4b, 4d, and 4f). However, the divergence for the India-Pakistan HW (Figure 4h) suggests that model-specific factors may influence the representation of the ACC's impact in this region.

In general, our results agree with previous studies quantifying the impact of ACC on HWs. For instance, in the case of the Pacific Northwest HW, Leach et al. (2024) estimated a contribution of 1.3°C (0.7, 1.6) for a global warming level of 1.2°C, whereas we obtain values between 0.9°C and 1.1°C (depending on the model, Figure 4b) for a lower warming level of 0.85°C (the one corresponding to the 1980–2014 period used to modify initial conditions). A similar estimate was obtained using PGW simulations with limited-area models: Bercos-Hickey et al. (2022) reported an ACC contribution of $0.98 \pm 0.40^\circ\text{C}$ in maximum temperatures using the WRF model, for a comparable global temperature delta imposed in the initial conditions. For the 2018 Iberian HW, our estimates (1.3°C–1.4°C, Figure 4d) match the findings of Barriopedro et al. (2020), who concluded that ACC amplified the HW by at least $>1^\circ\text{C}$. In the case of the Brazilian HW, Messori and Faranda (2024) reported a significant role of internal variability in driving this event, and a high degree of uncertainty, with the ACC contribution in daily maximum near-surface temperature ranging between 1°C and 4°C depending on the location—comparable to estimates from the World Weather Attribution (Kew et al., 2023). Our results indicate a weaker ACC signal of 1°C, although they are based on daily mean T850 averaged over a large region. Finally, for the early spring India-Pakistan HW, Zachariah et al. (2023) estimated a 1°C contribution for a global warming level of 1.2°C, which is comparable to the 0.5°C–0.8°C signal found here for a 0.85°C warming level (Figure 3h). A newly developed machine learning method based on convolutional neural networks also reported a similar ACC influence on the India-Pakistan HW magnitude (Trok et al., 2024).

Despite differences in event definitions, spatial averaging techniques, and climate baselines across the studies referenced above, our ACC signal estimates generally fall within the uncertainty ranges reported by those studies—highlighting the robustness of HW attribution results to methodological differences.

3.3. The Role of the Oceanic Forcing

In NWP-based attribution studies, it is common to modify SST boundary conditions along with the initial atmospheric conditions. This cannot be implemented explicitly in AIWP models (although the models may have learned to identify SST conditions from the initial atmospheric conditions). Ignoring the ACC signal coming from altered SSTs would yield partial attribution, and likely conservative estimates of ACC influences, assuming that SST changes add to the atmospheric signal. This question can be addressed with NeuralGCM, which allows to study of the influence of lower boundary forcing (SST and sea ice) on the attribution of the HWs analyzed here.

Figure 5 compares the ACC signal obtained from two distinct counterfactual experiments with NeuralGCM: one where only atmospheric thermodynamic variables (temperature and specific humidity) were modified, and another where only SST and sea ice cover were adjusted by removing the climate change signals present in global GCMs. The results show that much of the ACC signal in the weather forecasts originates from the initial state of atmospheric variables rather than boundary forcing. In most cases, the effect of ACC-induced SST changes on HW magnitude is weak. However, for the Brazilian and India-Pakistan HWs, a regional SST influence emerges, particularly over oceanic areas adjacent to the heat-affected regions. It suggests that, at short-to-medium lead times, atmospheric initial conditions alone are sufficient to capture most of the ACC-induced changes in HW intensity. This explains why AIWP models produce slightly lower but comparable ACC signals to NeuralGCM when SST and sea ice have also been modified to preindustrial conditions (shown in the previous subsection). For the Brazilian (Figure 5i) and India-Pakistan (Figure 5l) HWs, the impact of SST boundary forcing alone is small

Figure 4. Temporal evolution and uncertainty of the Anthropogenic climate change (ACC) signal during heatwaves (HWs) (a, b) 2021 Pacific Northwest HW; (c, d) 2018 Iberian HW; (e, f) 2023 Brazilian HW; and (g, h) 2022 India-Pakistan HW. (a, c, e, and g) Temporal evolution of the area-weighted ACC signal for daily mean T850 (°C) in the three models (see legend). Light shading represents the maximum and minimum values of the lagged ensemble (lead times of 1–5 days), while dark shading shows its interquartile range. (b, d, f, and h) Violin plots showing the probability distribution of the area-weighted ACC signal for T850 (°C) averaged over all days comprising the HW event for FourCastNet-v2 (blue), Pangu-Weather (red), and NeuralGCM (green). The whiskers span the maximum and minimum values of the lagged ensemble, with the mean in between (horizontal line); white dots indicate the median, and the boxes contain the interquartile range. Values displayed at the bottom of each violin correspond to the mean and the (5–95th)th interval. The black dashed lines show the area-averaged T850 delta imposed as part of the pseudo-global warming initial condition perturbation ($\Delta T_{850,ACC}$ in Figure 1). Blue boxes in Figure 3 are used to compute the area-averaged quantities.

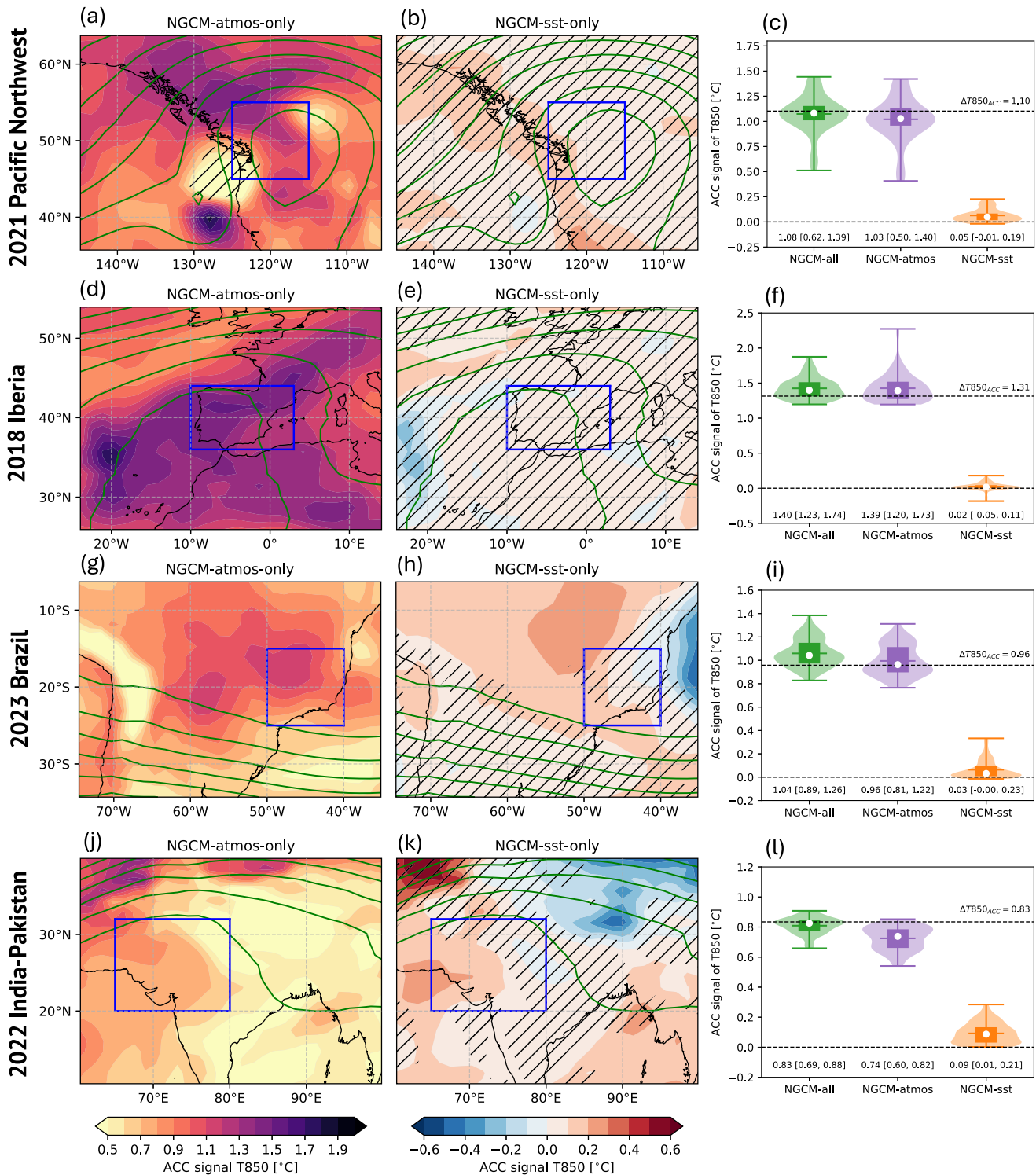


Figure 5. Influence of the oceanic boundary conditions on Anthropogenic climate change (ACC) attribution. ACC signal in T850 (°C) averaged over the heatwave (HW) duration, as simulated by NeuralGCM in counterfactual experiments with preindustrial (1850–1900) conditions in: (a, d, g, and j) atmospheric variables (temperature and specific humidity) only; (b, e, h, and k) boundary conditions (sea surface temperature and sea ice) only. (c, f, i, and l) Violin plots showing the probability distribution of the area-weighted ACC signal in T850 (°C) averaged over the HW duration for the three NeuralGCM counterfactual experiments with preindustrial conditions in the atmospheric and oceanic (green), atmospheric only (purple), and oceanic only (orange) variables. The whiskers span the maximum and minimum values of the lagged ensemble, with the mean in between (horizontal line); white dot indicate the median, and the boxes contain the interquartile range. Values at the bottom of each violin correspond to the mean and the (5–95)th interval. The blue boxes denote the areas used to compute the area-weighted averages. The black dashed lines in panels (c, f, i, and l) show the area-averaged T850 delta imposed as part of the pseudo-global warming initial condition perturbation ($\Delta T850_{ACC}$).

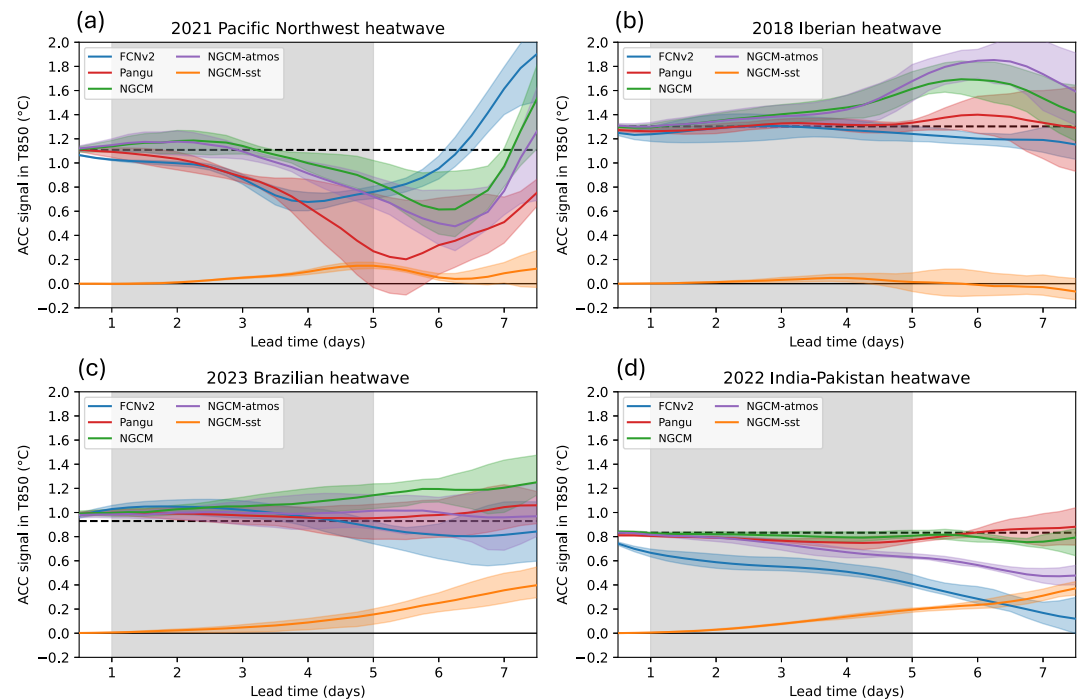


Figure 6. Lead time dependence of the Anthropogenic climate change (ACC) signal. Heatwave (HW) area-averaged T850 ACC signal as a function of the forecast lead time for (a) the 2021 Pacific Northwest HW, (b) the 2018 Iberian HW, (c) the 2023 Brazilian HW, and (d) the 2022 India-Pakistan HW. Colored lines represent different model experiments, with shading indicating the interquartile range (25th to 75th percentile) of daily values within each HW period. A 1-day centered running mean has been applied to smooth both the ensemble mean and the shading limits.

compared to the total atmospheric signal, but comparable in magnitude to the dynamical component, as seen from the difference relative to the black dashed line in the violin plots. Comparing the SST-forced spatial patterns in Figure 5 with the dynamically constrained atmospheric responses in Figure S6 in Supporting Information S1 further highlights the important role of SSTs in modulating the atmospheric response to the imposed thermodynamic perturbation.

A key feature of forecast-based attribution is that the ACC signal evolves with forecast lead time (Ermis et al., 2024; Leach et al., 2024), requiring a balance between capturing the event accurately and allowing sufficient time for the climate change signal to adapt to the specific weather situation. Figure 6 illustrates how the ACC signal evolves over lead time for each studied HW. At initialization, SST forcing has no direct impact on atmospheric temperature, as inferred from the comparison of the two NeuralGCM experiments. However, for longer lead times, interactions with the ocean become increasingly important, gradually influencing the ACC signal. This effect is particularly pronounced in the 2023 Brazilian and 2022 India-Pakistan HWs (Figures 6c and 6d), for which the ACC signal steadily strengthens over time. Nevertheless, the ACC signal driven purely by boundary forcing remains weak—approximately an order of magnitude smaller than the signal from initial conditions—particularly for lead times shorter than 5 days, which are the ones used for the conditioned attribution. Since forecast attribution primarily relies on these lead times, the contribution from boundary forcing to ACC signal on HWs can be neglected.

For the Brazilian HW, the ACC signal predicted by Pangu-Weather closely matches the NeuralGCM simulation without SST forcing (purple line in Figure 6c), whereas for the India-Pakistan event, it aligns more with the NeuralGCM simulation in which both SSTs and atmospheric initial conditions are modified (green line in Figure 6d). This indicates that Pangu-Weather is able to capture the indirect influence of SSTs in the India-Pakistan case, but not in the Brazilian one. In contrast, FourCastNet-v2 displays a consistent cooling drift for both events, suggesting that it cannot represent the effects of the SST forcing on the HW magnitude. For the other two HWs, that is, Pacific Northwest and Iberian, where SSTs appear to play a minimal role, the discrepancies between the AIWP models and NeuralGCM are likely due to the intrinsic chaotic nature of atmospheric

circulation and different model biases. This interpretation is supported by the growing spread among model predictions with increasing lead time, particularly pronounced for the Pacific Northwest HW (Figure 6a), where predictability is especially limited (Figure 2c). For the Iberian HW (Figure 6b), the temperature ACC signal increases with forecast lead time, a trend that is particularly evident in both NeuralGCM simulations (with and without SST forcing). This observation implies that the atmospheric circulation pattern may be amplifying the ACC signal in this case, as suggested by previous results.

To sum up, while our results reinforce the idea that SST and sea ice can modulate extreme heat events, their influence at the timescales considered in this study is secondary to that of atmospheric initial conditions. The ACC signal seen in short-term forecasts is primarily driven by direct atmospheric warming, rather than sustained interactions with the ocean. This highlights the dominant role of atmospheric thermodynamic changes in shaping HW events, with oceanic forcing playing a slowly varying and regionally dependent role.

4. Discussion and Conclusions

This article presents a hybrid approach for the attribution of extreme events to ACC by combining AI weather forecasts with physics-based climate information from GCMs. The attribution is performed by comparing AI-based weather forecasts from the factual world, where the event unfolds, with those from a counterfactual world, in which human-induced changes in the atmospheric conditions are removed in a physically consistent manner by using the output from GCM simulations. This methodology integrates a climate-weather perspective and offers several advantages over similar approaches applied to NWP models. First, it enables global-scale predictions and near-real-time attribution of multiple events simultaneously. Being computationally efficient and widely accessible, this approach addresses the need for timely attribution (Stott & Christidis, 2023) by evaluating ACC influence on forecasted rather than solely observed outcomes. Finally, the method quantifies the impact of climate change on instantaneous and spatially resolved fields, capturing changes in event's characteristics beyond intensity (e.g., spatial evolution, size, etc.). This also avoids some of the “framing” constraints of traditional methods, such as the predefined definition of the event, which can influence attribution results (NAS, 2016).

The hybrid attribution approach is applied to four historical HWs with different characteristics across various regions and seasons, using three AI-based forecasting models: FourCastNet-v2, Pangu-Weather, and NeuralGCM (a hybrid AI-physics model). Overall, all three models perform well in predicting the diverse spatial patterns and intensities of these HWs at short lead times (1–5 days). Some events, however, are better predicted than others. For example, the magnitude of the 2021 Pacific Northwest HW is notably underestimated by FourCastNet-v2, illustrating the forecasting challenges posed by unprecedented events (black swans) and the rapid error growth associated with midlatitude dynamics. Pangu-Weather and NeuralGCM also struggle to capture the intensity of this event, but at longer forecasting lead times (>5 days). In contrast, for the 2018 Iberian, 2022 India–Pakistan, and 2023 Brazilian HWs, the models generally exhibit low area-averaged biases (<1°C). NeuralGCM often produces the most accurate forecasts, though not by a large margin, nor in all cases analyzed (Figure 2). In many instances, Pangu-Weather and FourCastNet-v2 perform similarly to, and in one event, even outperform NeuralGCM's forecasts. These findings support the use of AIWP models as practical tools for short-term HW forecasting.

The ACC signals—computed as the difference between factual and counterfactual forecasts—show a broadly consistent response across different models for all events analyzed (Figure 3). While much of the ACC signal is already present in the initial condition perturbation (Figure S6 in Supporting Information S1), the AIWP models modulate this thermodynamic signal in a way that reflects the specific synoptic evolution of each event. As a result, the intensity and spatio-temporal pattern of the attributable signal varies across events, influenced by both the imposed perturbation and the atmospheric circulation at initialization. For the Iberian and Brazilian HWs, the ACC signals in T850 resemble a localized amplification pattern that coincides with anticyclonic conditions. For the Pacific Northwest and Indian HWs, however, the attributable T850 spatial pattern is more complex. In particular, part of the ACC signal in the Pacific Northwest event appears to stem from a growing model bias near the coast (Figure S2 in Supporting Information S1 and Figures 3a–3c). However, this link between biases and ACC signals doesn't extend to the other HWs. For instance, during the India–Pakistan HW, all models exhibit similar biases, yet FourCastNet-v2 shows a weaker ACC signal compared to the other two models. Consequently, we do not find a consistent relationship between forecasting bias and the attributable ACC signal across the HWs

analyzed (Figure S11 in Supporting Information S1). Although previous studies have warned that unreliable climate simulations may lead to overestimated attribution signals (Bellprat & Doblas-Reyes, 2016), these issues in climate-oriented attribution are not expected to affect forecasts constrained by initial conditions.

The attribution signals identified in our short-term forecasts are primarily shaped by atmospheric initial conditions rather than boundary forcing from SST or sea ice (Figure 5). However, SSTs can play a non-negligible role in modulating the dynamically constrained response, particularly during certain events. For instance, in the Brazilian and India–Pakistan HWs, the contribution of SST forcing is comparable to the dynamically integrated atmospheric response (Figure 5 and Figure S6 in Supporting Information S1). In contrast, the effect of SSTs is minor for the Iberian and Pacific Northwest HWs. These findings support the use of AIWP models for short-term attribution of HWs, where forecasts are conditioned on the observed synoptic evolution and the ACC signal primarily reflects the imposed thermodynamic perturbation. However, for certain extreme events—such as (extra-)tropical cyclones—accurate attribution may require AIWP models that explicitly account for SST variability, such as NeuralGCM. Yet recent work suggests that NeuralGCM may underestimate the ACC signal in HWs compared to Earth system models, due to missing land–atmosphere feedbacks, such as soil moisture–temperature coupling, which play a key role in amplifying extremes over land (Duan et al., 2025). Future work should therefore explore the development and testing of next-generation AIWP models that explicitly account for soil properties and land–atmosphere coupling—such as the ECMWF data-driven model AIFS (Lang, Alexe, Chantry, et al., 2024)—thereby improving the realism and robustness of attribution in regions where surface feedbacks are critical.

An additional direction for future research is to explore the direct effect of radiative forcing from atmospheric greenhouse gases, which is missing in the models employed here but not in NWP-based attribution studies (Ermis et al., 2024; González-Alemán et al., 2023; Leach et al., 2024). Newly available AI-based models (Duncan et al., 2024; Watt-Meyer et al., 2025) with explicit specification of greenhouse gas concentrations offer a promising opportunity to investigate this impact on the attribution estimates by performing similar factorial experiments to those presented herein.

A potential limitation of AIWP models is their ability to handle conditions that lie outside the distribution of their training period. However, recent studies indicate that these models generalize well to both historical and future climates beyond their training period (Rackow et al., 2024), supporting their use for extreme event attribution under altered climate conditions. Despite this, model drift toward the training climatology can occur when simulations are initialized in years substantially warmer or colder than the training baseline (1979–2017). In particular, the HW events analyzed here occurred in years 0.4°C–0.7°C warmer than the training mean, while counterfactual conditions are roughly 0.85°C colder than factual ones, thus placing them within the training distribution. Notably, in extreme years like 2023, the counterfactual climate aligns more closely with the training data than the factual conditions. This setup reduces concerns about out-of-distribution generalization and model instability in the counterfactual experiments.

In our experiments, factual simulations exhibit a drift in global mean T850 of up to 0.2°C over five forecast days for models such as FourCastNet and Pangu-Weather, while counterfactual simulations show weaker or negligible drifts, typically below 0.1°C (Figure S12 in Supporting Information S1). These findings are consistent with those of (Rackow et al., 2024), who reported a cooling trend in Pangu-Weather, also when running under colder climates. This systematic drift reduces the ACC signal by about 0.1°C over the 1–5 days lead time window used in our methodology (Figure S13 in Supporting Information S1). A similar but more pronounced effect is observed for FourCastNet, due to a cooling bias in factual runs and a slight warming trend in counterfactual simulations. In contrast, NeuralGCM shows negligible drift across all events, which may explain why it consistently produces the largest ACC signals in our attribution experiments (Figure 4). Although the spatial and temporal structure of drift might vary substantially with initialization state and model, its overall impact on the magnitude of attribution remains relatively small. Overall, the attribution signals are consistent across all models and events, confirming the robustness of the approach despite background climate shifts and model-specific biases.

We quantify the uncertainty in event predictability using a lagged ensemble approach (Brenowitz et al., 2024). Future work could explore the use of probabilistic forecasts generated from an ensemble of perturbed initial conditions, like in recently launched models such as GenCast (Price et al., 2025), the AIFS-CRPS ensemble prediction model (Lang, Alexe, Clare, et al., 2024), or the stochastic version of NeuralGCM (Kochkov et al., 2024). However, uncertainty due to imperfect knowledge of the initial conditions is expected to be smaller

than uncertainty stemming from differences in initialization times. As a result, the uncertainty of the ACC signal is likely overestimated—especially at the short lead times considered here. The attribution signal is inherently sensitive to lead time, since this determines both the extent to which the forecast remains constrained to the observed synoptic state and the time available for the imposed ACC perturbation to influence the atmospheric evolution. Short lead times offer stronger dynamical constraint but limited signal emergence, while longer lead times allow more thermodynamic adjustment at the cost of increasing uncertainty. This trade-off is acknowledged in forecast-based attribution (Ermis et al., 2024; Leach et al., 2024), underscoring the importance of careful lead time selection. In nudged storyline simulations, by contrast, the sensitivity to lead time is largely eliminated because the large-scale flow is continuously constrained toward reanalysis throughout the simulation (Klimiuk et al., 2025; Sánchez-Benítez et al., 2022; van Garderen et al., 2021). However, this approach requires the full event to have already unfolded and thus cannot support real-time or forward-looking attribution. In our case, where forecasts evolve freely from perturbed initial conditions, the lagged ensemble approach offers a practical means to explore lead-time sensitivity and assess the robustness of attribution signals across a synoptically constrained window.

Although our forecasts are initialized with ERA5 reanalysis, which has a 5-day delay, real-time forecasts using the Integrated Forecast System (IFS) atmospheric analysis would allow attribution before HWs occur. In fact, we do not find substantial differences between the attribution results obtained from ERA5 and the IFS analysis (Figure S14 in Supporting Information S1), underscoring the feasibility of anticipatory global attribution.

While this study focuses on HWs, the same approach can be applied to other extreme weather events, such as tropical and extratropical cyclones, which are also influenced by ACC (Catto et al., 2019; Knutson et al., 2020; Priestley & Catto, 2022; Reed et al., 2022). For these extreme events, AIWP models capture physical processes realistically (Hakim & Masanam, 2024) and exhibit consistent sensitivities to initial condition perturbations (Baño-Medina et al., 2025). However, challenges remain, particularly in preserving key atmospheric balances and resolving sub-synoptic scales (Bonavita, 2024; Selz & Craig, 2023), which are crucial for accurately attributing small-scale events like tropical storms and extreme convective precipitation. To further assess the reliability of AIWP models, here we compared them with a hybrid model that explicitly resolves atmospheric dynamics. The similarity in their forecasts and ACC signals suggests that AIWP models capture physically consistent responses to initial condition changes and reinforces the potential of AIWP models for the attribution of other extreme events.

In summary, this work highlights the potential of AIWP models to speed up the process of attributing individual extreme events to climate change. By combining AI-driven forecasts initialized with climate signals from global climate simulations, we show that it is possible to assess the role of ACC in extreme heat events on a global scale, using far less computing power than traditional NWP models, and with results that are similar to those from physically informed models. This approach opens the door to operational, anticipatory attribution—where the influence of human-made climate change on extreme events can be quantified even before an event unfolds. This AI-driven approach could also offer a valuable tool for decision-making (e.g., the provision of scientific evidence for prompt activation of Loss and Damage funds, Coumou et al., 2024).

Data Availability Statement

ERA5 reanalysis data (Hersbach et al., 2020), including hourly single-level and pressure-level fields, was obtained from the Copernicus Climate Data Store (Copernicus Climate Change Service, 2023a, 2023b) using the CDS API. The training weights and code for running FourCastNet-v2 and Pangu-Weather were obtained using the Open Source Earth2MIP package (NVIDIA, 2023). The NeuralGCM model code (version 1.1.2) is publicly available on Zenodo (Hoyer et al., 2025), with source code hosted at <https://github.com/neuralgcm/neuralgcm>. CMIP6 historical simulation data (Eyring et al., 2016) were obtained from the Earth System Grid Federation (ESGF) data portals (World Climate Research Programme, 2019).

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